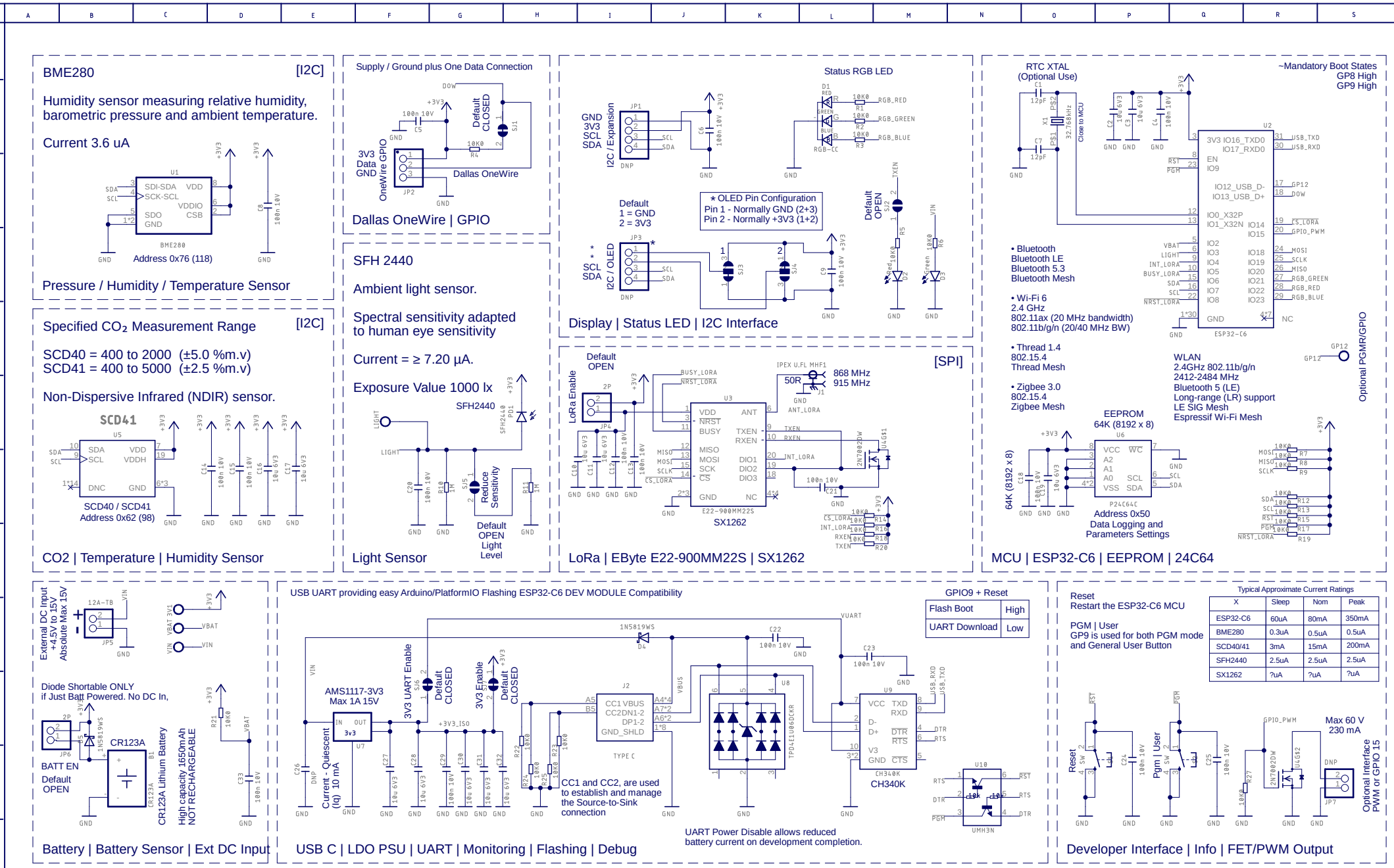
BME280
[I2C]
Humidity sensor measuring relative humidity, barometric pressure and ambient temperature.
Current 3.6 uA
Address 0x76 (118)
Pressure / Humidity / Temperature Sensor
Supply / Ground plus One Data Connection
Default CLOSED
3V3 Data GND
OneWire GPIO
Dallas OneWire
Dallas OneWire | GPIO
SFH 2440
Ambient light sensor.
Spectral sensitivity adapted to human eye sensitivity
Current = ≥ 7.20 µA.
Exposure Value 1000 lx
Reduce Sensitivity
Default OPEN Light Level
Light Sensor
Status RGB LED
GND 3V3 SCL SDA
I2C / Expansion
Default 1 = GND 2 = 3V3
I2C / OLED
* OLED Pin Configuration
Pin 1 - Normally GND (2+3)
Pin 2 - Normally +3V3 (1+2)
Default OPEN
Display | Status LED | I2C Interface
RTC XTAL (Optional Use)
~Mandatory Boot States
GP8 High
GP9 High
• Bluetooth
Bluetooth LE
Bluetooth 5.3
Bluetooth Mesh
• Wi-Fi 6
2.4 GHz
802.11ax (20 MHz bandwidth)
802.11b/g/n (20/40 MHz BW)
• Thread 1.4
802.15.4
Thread Mesh
• Zigbee 3.0
802.15.4
Zigbee Mesh
WLAN
2.4GHz 802.11b/g/n
2412-2484 MHz
Bluetooth 5 (LE)
Long-range (LR) support
LE SIG Mesh
Espressif Wi-Fi Mesh
Optional PGMR/GPIO
EEPROM 64K (8192 x 8)
64K (8192 x 8)
Address 0x50
Data Logging and Parameters Settings
MCU | ESP32-C6 | EEPROM | 24C64
Specified CO₂ Measurement Range
[I2C]
SCD40 = 400 to 2000 (±5.0 %m.v)
SCD41 = 400 to 5000 (±2.5 %m.v)
Non-Dispersive Infrared (NDIR) sensor.
SCD40 / SCD41
Address 0x62 (98)
CO2 | Temperature | Humidity Sensor
Default OPEN
LoRa Enable
[SPI]
868 MHz
915 MHz
SX1262
LoRa | EByte E22-900MM22S | SX1262
External DC Input +4.5V to 15V Absolute Max 15V
Diode Shortable ONLY if Just Batt Powered. No DC In,
BATT EN
Default OPEN
CR123A
CR123A Lithium Battery
High capacity 1650mAh NOT RECHARGEABLE
Battery | Battery Sensor | Ext DC Input
USB UART providing easy Arduino/PlatformIO Flashing ESP32-C6 DEV MODULE Compatibility
AMS1117-3V3 Max 1A 15V
3V3 UART Enable
3V3 Enable
Current - Quiescent (Iq) 10 mA
CC1 and CC2, are used to establish and manage the Source-to-Sink connection
CH340K
UART Power Disable allows reduced battery current on development completion.
USB C | LDO PSU | UART | Monitoring | Flashing | Debug
GPIO9 + Reset
Flash Boot High
UART Download Low
Reset
Restart the ESP32-C6 MCU
PGM | User
GP9 is used for both PGM mode and General User Button
Typical Approximate Current Ratings
X Sleep Nom Peak
ESP32-C6 60uA 80mA 350mA
BME280 0.3uA 0.5uA 0.5uA
SCD40/41 3mA 15mA 200mA
SFH2440 2.5uA 2.5uA 2.5uA
SX1262 ?uA ?uA ?uA
Max 60 V 230 mA
Optional Interface PWM or GPIO 15
Developer Interface | Info | FET/PWM Output
DitroniX
Home Automation | Internet of Things | Energy Monitoring
Production
(C)2021-2026 DitroniX - Dave Williams G8PUO
Main PCB 63x51
ESP32-C5 SX1262 SDK
ditronix.net
DitroniX Ltd – UK
AQIL ESP32C6 LoRa SCD41 SDK v1.2603.102
AIR-QUALITY-MONITOR-IOT-WEATHER-STATION-LORA
Sheet 1/1
Drwg/Chk: Dave Williams
12/03/2026 16:17